

Privacy-Preserving Tucker Train Decomposition Over Blockchain-Based Encrypted Industrial IoT Data

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I. INTRODUCTION

Abstract—Tucker decomposition has been widely used to extract meaningful and underlying data from heterogeneous data generated by different kinds of devices in a wide range of industrial Internet of Things (IIoT) applications. IIoT data uploaded to the cloud contain personal and sensitive information; thus, there is a growing concern about data privacy. Current existing data analysis solutions, however, assume that the data are reliably and securely collected from different IIoT data providers, an assumption that is not always true in the real world. To address the issues, in this article we propose a privacy-preserving tucker train decomposition based on gradient descent over blockchain-based encrypted IIoT data. Specifically, we use blockchain techniques to enable IIoT data providers to reliably and securely share their data by encrypting them locally before recording them in the blockchain. We use tensor train (TT) theory to build an efficient TT-based tucker decomposition based on gradient descent that tremendously reduces the number of elements to be updated during the tucker decomposition. We utilize the massive resources of fogs and clouds to implement an efficient privacy-preserving tucker train decomposition scheme. We use homomorphic encryption to build our scheme that does complete tucker train decomposition without the involvement of users. Results from a series of extensive experiments on synthetic datasets and real-world datasets demonstrate that our proposed scheme is efficient.

Index Terms—Blockchain, cloud computing, encrypted data, industrial IoT (IIoT), privacy protection, tensor, tensor factorization.

THE LAST decades have witnessed an exponential growth of data as a result of the ever-increasing cutting-edge industrial IoT (IIoT) infrastructures in different application domains, such as smart homes, smart transportation, and smart cities [1], [2]. The enormous volume of data has ignited extensive research both in industry and academia on how to utilize the data to enhance industrial services provided to users for better and enhanced living. IIoT has been adopted in many application areas to help bring better service closer to the end-users. In health care, for example, personal fitness information recorded by wearable sensors can be used partly in tucker decomposition for accurate diagnosis. IIoT systems are distributed, real-time, adaptive, and intelligent systems involving complex heterogeneous data from hyperspaces. The main focus of IIoT is to integrate and coordinate all data collected from cyber, and physical spaces to extract useful information used to provide high-quality personalized services to people as well as optimizing industrial processes [3], [4].

Smaller samples of data that lack in volume and variety are usually distributed over different geographical locations, i.e. in hospitals or smart cities. For efficient feature representations, however, a large data sample is required. Collecting these smaller datasets under the current system structure not only poses privacy concerns but also has a high communication overhead that often prevents the data from being shared. A mechanism for efficiently aggregating the smaller datasets from different data providers has long been overdue in IIoT.

The data collected by IIoT devices are mostly sensitive since they are personal and private data. This, therefore, raises a number of data integrity, privacy, and ownership concerns as the data are shared across many participants [5], [6]. For example, sensitive and confidential data may be leaked while being processed, tampered with, or modified while being shared or aggregated, and finally, IIoT data providers may no longer have control of their data since the shared data can be accessed by many parties, thus enabling it to be replicated by others without their permission. As a result of the aforementioned data integrity, privacy, and ownership concerns, many data providers have been reluctant to share their data for processing [7].

Data privacy concerns in data analysis have witnessed extensive research in both government, industry, and academia. Current researches have resorted to either using differential

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privacy (DP) or using cryptographic algorithms in protection of data privacy for individual IIoT data providers [8], [9]. These techniques, however, assume that data can reliably be provided for subsequent processing and thus takes little attention in data integrity and ownership. Furthermore, these mechanisms are theoretical but currently very expensive to implement. To address this assumption, we use blockchain techniques [10] to build a secure decentralized data-sharing platform.

Blockchain is basically an immutable distributed ledger. Owing to its interesting features, such as coordination, record keeping, and irrevocable transactions, which are fundamental in society, blockchain can thus work as a public records repository for entire regions, being able to include all documents, assets, identities, and events. The blockchain was purposely designed to make it possible to securely share data among many data providers without the need of a centralized trusted third party [10].

Even though blockchain has a huge potential going forward, there are currently challenges on how to incorporate tucker decomposition in the blockchain. One evident challenge is how to design an appropriate data format for highly dimensional data that can easily be accommodated on the blockchain while protecting data privacy for the individual data providers. Another challenge is how to construct an accurate tucker decomposition scheme for blockchain data that can protect the privacy of sensitive information.

To address these issues, we propose a privacy-preserving tucker train decomposition (PPTTD) based on gradient descent over blockchain-based encrypted IIoT data. The proposed scheme uses homomorphic encryption to protect data privacy in IIoT environments. The additive homomorphic cryptosystem has a higher efficiency compared to other algorithms when considering the efficiency in terms of encryption and decryption.

Traditional tucker decomposition optimization algorithms on plain text needs to update many elements in every iteration; thus, they would result in many interactions between data providers and data analysts if they are applied directly on encrypted data. To minimize the interactions required in processing encrypted data, we use gradient descent and tensor train (TT) to construct secure building blocks with homomorphic properties. By using these building blocks to implement our proposed scheme, the number of interactions is significantly reduced, thus a reduction in the overall computation and communication overhead.

Specifically, the main contributions of this article are three fold, which are as follows.

- 1) We use blockchain techniques to realize a reliable and secure IIoT data-sharing platform. IIoT data providers encrypt their corresponding IIoT data from IIoT devices in their domain locally before they are recorded in the blockchain using special formatted transactions.
- 2) We use TT theory to build an efficient TT-based tucker decomposition based on gradient descent that tremendously reduces the number of elements that need to be updated during the gradient-descent tucker decomposition. We utilize the massive resources of fogs and clouds to present a privacy-preserving computing model, based on which we implement an efficient privacy-preserving TT-based

tucker decomposition scheme. The scheme does complete tucker decomposition without the involvement of users and trusted third party by employing homomorphic encryption.

- 3) We perform extensive experiments on synthetic datasets and real-world datasets for validating that our scheme can securely and efficiently perform tucker decomposition with high accuracy.

The rest of this article is organized as follows. Section II presents preliminaries. Thereafter, Sections III and IV present the proposed method and evaluation of experiment, respectively. Section V presents a summary of the of related work. Finally, Section VI concludes this article.

II. PRELIMINARIES

A. Blockchain Systems

A blockchain system is basically an immutable distributed ledger consisting of continuously growing append only records [10], [11]. Such records include payment transactions, contractual agreements, product sales, etc. Blockchain was essentially designed to support reliable transactions for untrusted parties making a transaction. Owing to its disruptive effect, many kinds of cross-industry open platforms have been proposed, i.e., Ethereum, Ripple, and HyperLedger, and have seen their applications in different IIoT scenarios. With regard to access control, blockchains can be classified as either private, consortium, or public. Blockchain has interesting features, which are fundamental in society. The following are a few of such features that make blockchain fit for holding data.

1) *Decentralization*: Blockchain as a distributed ledger consists of distributed nodes set in a peer-to-peer network, with each node (peer) maintaining a copy of the blockchain, thus, no trusted or central administration is required. The participating peers (nodes) validate and relay all transactions in the blockchain. The data replication in different nodes also eliminates the single point of failure, which are a major problem in centralized systems.

2) *Tamper-Proof*: Blockchain utilizes consensus mechanism to control the addition of new blocks. The consensus mechanism, i.e., proof of work (PoW), discourages dishonest parties by making it difficult to perform transactions as a result of, for example, the computational power needed to accomplish the task. The nature of PoW is, therefore, to impose computational overhead in computing and validating transactions to inhibit block alteration making it practically impossible to manipulate blockchain transactions.

3) *Traceability*: Transaction data in a blockchain are easily verifiable by any authorized party as all transactions are visible to the public. It is thus possible to track a transaction down to its source while data owners can enjoy their benefits in real time since they can easily get paid as their data are being used by third parties.

B. Tucker Decomposition

Tucker decomposition is a mechanism for expressing a tensor using fewer feature spaces for dimensionality reduction [5], [12].

By tucker decomposition, a third-order tensor $\underline{T} \in R^{I_1 \times I_2 \times I_3}$ (to make a distinction with the matrix, we use underlined uppercase letters to represent the tensor, e.g., $\underline{T}$) can be decomposed into a core tensor and factor matrices equaling to the number of modes of the tensor, as shown in (1). From (1), $U_1 \in R^{I_1 \times R_1}$, $U_2 \in R^{I_2 \times R_2}$, and $U_3 \in R^{I_3 \times R_3}$ are factor matrices on the three different modes, whereas $\underline{C} \in R^{R_1 \times R_2 \times R_3}$ is the core tensor

$$\underline{T} = \underline{C} \times_1 U_1 \times_2 U_2 \times_3 U_3. \quad (1)$$

Tucker decomposition is not exact but rather yields the best approximation. The loss resulting from this approximation can be defined as a loss function in the form

$$L(U_1, U_2, U_3, \underline{C}) = \frac{1}{2} \|\underline{T} - \underline{C} \times_1 U_1 \times_2 U_2 \times_3 U_3\|_F^2. \quad (2)$$

In each iteration of the tucker decomposition, the aforementioned loss function is minimized until it reaches its minimum value. When the value of the loss function is at its minimum, then U_1 , U_2 , U_3 , and $\underline{C}$ satisfies $\underline{T} \approx \underline{C} \times_1 U_1 \times_2 U_2 \times_3 U_3$. The method based on gradient descent can be used to implement tucker decomposition.

C. TT Decomposition

TT decomposition is a new mechanism for approximating a tensor [13], [14]. Relative to tucker decomposition, TT decomposition can express higher order approximate tensors with fewer parameters to avoid exponential growth of parameters with order [15], [16]. Relative to the CANDECOMP/PARAFAC (CP) decomposition, TT decomposition is a stable form of expression, which avoids the NP-hard problem of “optimal rank solution” in CP decomposition and the problem that CP decomposition easily falls into local minima. In addition, the TT decomposition is based entirely on the QR and singular value decomposition (SVD) decompositions of a series of matrices, without any recursion. Therefore, TT decomposition is widely used in many fields.

TT decomposition is the process of expressing a tensor $X \in R^{I_1 \times I_2 \times \dots \times I_N}$ into a set of low-order tensor cores, i.e., $G_1^X, G_2^X, \dots, G_N^X$, forming a reduced tensor expression as in the following:

$$X \approx G_1^X \times_1 G_2^X \times_2 \dots \times_{n-1} G_N^X \quad (3)$$

where the size of tensor G_n^X is $RA_{n-1} \times I_n \times RA_n$ for $n = 1, 2, \dots, N$, $RA_0 = RA_N = 1$.

III. CONSTRUCTION OF PPTTD

In this section, we present in detail our proposed privacy-preserving tucker decomposition based on TT over blockchain-based encrypted IIoT data.

A. System Overview

Our framework entails building a PPTTD on top of the blockchain. The scheme uses data stored in the blockchain, which are formatted and encrypted by the IIoT data providers before being saved.

An encrypted tensor is constructed from the data stored in the blockchain. Using the building blocks, e.g., secure subtraction,

secure tensor outer product, and secure mode-n tensor product, we construct our privacy-preserving tucker decomposition based on TT that takes the encrypted tensor as input.

We conceptualize a data-driven IIoT platform as follows, which comprises IIoT devices, IIoT data providers, cloud-enabled blockchain-based IIoT platform, and privacy-preserving tucker decomposition module.

- 1) IIoT devices can sense and transmit important data from IIoT ecosystems. The data come from a wide range of applications, e.g., smart cities and smart transport systems. It is worth noting that IIoT devices have limited storage and computing power, and thus cannot be used for heavy computations.
- 2) IIoT data providers are responsible for collecting valuable data from all the IIoT devices in their domain. Data from most of the IIoT ecosystems are sensitive. Therefore, each IIoT data provider must encrypt its data locally using homomorphic encryption before recording them in the blockchain.
- 3) Cloud-enabled blockchain-based IIoT platform is responsible for storing the encrypted IIoT data from all the IIoT data providers. The built-in features of the blockchain make the data stored in the chain to be tamper-proof, immutable, and authentic, thus can be considered as a single source of truth. The platform also makes it possible for multiple data providers to securely share their data.
- 4) Tensor decomposition module built on top of blockchain performs insightful analysis using the data stored in the cloud-enabled blockchain. By performing the secure tucker decomposition based on TT, underlying interesting features of the stored IIoT data can be discovered and used for predicting and offering better service to the IIoT users.

The aim of our framework is to protect the privacy of participating end-users and be able to perform tucker decomposition securely in IIoT. Specifically, each IIoT data provider's privacy is the IIoT records in their domain. In this regard, we formulate following two security goals.

- 1) Protection of the privacy of individual IIoT data providers if confronted by honest but curious adversaries.
- 2) Protection of the privacy of individual IIoT data providers if one or more of them collude with IIoT data analysts.

B. Encrypted Data Sharing via Blockchain

In this section, we discuss in detail the process of sharing data through the blockchain. To get accurate results in our tucker decomposition step, we assume that the data are processed and effectively represented as a tensor. For data to be stored in the blockchain, they have to be formatted into a transaction. Unlike normal blockchain transactions, our transaction is divided into two parts, labeled input and output fields. The input field records the IIoT device type, IIoT data provider address, and encrypted data while its corresponding output field consists of IIoT device type, encrypted data, and address of the tucker decomposition function. Once an IIoT data provider encrypts the IIoT data, the provider sends the encrypted data to the blockchain for storage, with the blockchain node receiving the transaction, broadcasting

Algorithm 1: Tucker Decomposition Based on Tensor Train.

Input: Tensor $\underline{T} \in R^{I_1 \times I_2 \times \dots \times I_N}$, maximum number of iterations $Iter_{\max}$, error threshold ε ;
 The dimensions $R_1, R_2, \dots, R_N$ of core tensor $\underline{C}$, number $c_1, c_2, \dots, c_N$ of tensor train cores;
 Size $I_{11} \times I_{12} \times \dots \times I_{1(c_1-1)} \times I_{21} \times I_{22} \times \dots \times I_{2(c_2-1)} \times \dots \times I_{N1} \times I_{N2} \times \dots \times I_{N(c_N-1)}$ transformation for tensor $\underline{T}$, ranks $r_{11}, r_{12}, \dots, r_{1(c_1-1)}, r_{21}, r_{22}, \dots, r_{2(c_2-1)}, \dots, r_{N1}, r_{N2}, \dots, r_{N(c_N-1)}$ of tensor train cores.
Output: Core tensor $\underline{C} \in R^{R_1 \times R_2 \times \dots \times R_N}$, N tensor trains $\bar{U}_1, \bar{U}_2, \dots, \bar{U}_N$, where every tensor train $\bar{U}_k = U_k^{(1)} \times_1 U_k^{(2)} \times_k \dots \times_1 U_k^{(c_k)}$.

- 1: Transform tensor $\underline{T}$ to get tensor T according to dimensions;
- 2: Initialize the tensor train cores $U_k^{(1)} \in R^{1 \times R_k \times r_{k1}}; U_k^{(j)} \in R^{r_{k(j-1)} \times I_{k(j-1)} \times r_{kj}}, 1 < j < c_k; U_k^{(c_k)} \in R^{r_{k(c_k-1)} \times I_{k(c_k-1)} \times 1}; j < c_k$;
- 3: Initialize $\underline{C} \in R^{R_1 \times R_2 \times \dots \times R_N}$;
- 4: Set step size η ;
- 5: **while** $t < Iter_{\max}$ and $L_t > \varepsilon$ **do**
- 6: **for each** $t_{i_{11}i_{12}\dots i_{1(c_1-1)}i_{21}i_{22}\dots i_{2(c_2-1)}\dots i_{N1}i_{N2}\dots i_{N(c_N-1)}}$ **do**
- 7: Compute $\nabla_{\underline{C}} L$ by Equation 4;
- 8: Compute $\nabla_{U_k^{(j)}} L$ for each core $U_k^{(j)} (k = 1, 2, \dots, N, 0 < j \leq c_k)$ by Equation 5;
- 9: $\underline{C} = \underline{C} - \eta \nabla_{\underline{C}} L$;
- 10: $U_k^{(j)} = U_k^{(j)} - \eta \nabla_{U_k^{(j)}} L$ for $k = 1, 2, \dots, N, 0 < j \leq c_k$;
- 11: $t = t + 1$;
- 12: **end for**
- 13: **end while**

it to the entire peer to peer (P2P) network for validation. Peers verify the correctness of incoming transactions and package them into a block. Once a node successfully mines a block, the node broadcasts it to all other nodes for verification. Blockchain nodes use consensus mechanisms to validate new blocks, after which it is appended to the blockchain once found valid. It is worth noting that a block may include several transactions.

C. Building Blocks

1) Tucker Decomposition Based on TT: We present an efficient tucker decomposition algorithm based on TT and gradient descent.

The main idea behind the tucker decomposition algorithm based on TT is the following. Using transformation and TT theory, the objective function on the core tensor $\underline{C}$ and the matrix U_n is equivalently transformed into the objective function on the core tensor $\underline{C}$ and the TT cores of the tensors, which are gotten by reshaping the matrix U_n . The number of parameters in gradient calculations is reduced. The main steps involved in the proposed approach are highlighted in Algorithm 1.

First, a higher order tensor T is gotten by transforming the tensor $\underline{T}$ according to the dimensions $I_{11}, I_{12}, \dots, I_{1(c_1-1)}, I_{21}, I_{22}, \dots, I_{2(c_2-1)}, \dots, I_{N1}, I_{N2}, \dots, I_{N(c_N-1)}$ in step 1.

Then, the TT cores $U_k^{(1)} \in R^{1 \times R_k \times r_{k1}}; U_k^{(j)} \in R^{r_{k(j-1)} \times I_{k(j-1)} \times r_{kj}}, 1 < j < c_k; U_k^{(c_k)} \in R^{r_{k(c_k-1)} \times I_{k(c_k-1)} \times 1}$, and $\underline{C} \in R^{R_1 \times R_2 \times \dots \times R_N}$ are initialized. The step size η is set.

When $t < Iter_{\max}$ and $L_t > \varepsilon$, fog and cloud perform the following steps. For each $t_{i_{11}i_{12}\dots i_{1(c_1-1)}i_{21}i_{22}\dots i_{2(c_2-1)}\dots i_{N1}i_{N2}\dots i_{N(c_N-1)}}$, $\nabla_{\underline{C}} L$ is computed, $\nabla_{U_k^{(j)}} L$ is computed for each core

$U_k^{(j)} (k = 1, 2, \dots, N, 0 < j \leq c_k)$, $\underline{C} = \underline{C} - \eta \nabla_{\underline{C}} L$ is gotten, and $U_k^{(j)} = U_k^{(j)} - \eta \nabla_{U_k^{(j)}} L$ is gotten for $k = 1, 2, \dots, N, 0 < j \leq c_k$.

The gradients for variables $U_k^{(j)}$, $\underline{C}$ are derived as follows:

$$\begin{aligned} \nabla_{\underline{C}} L = & \left(\underline{C} \times_1^1 \left(U_1^{(1)} \times_1 U_{1(i_{11}:)}^{(2)} \times_1 U_{1(i_{12}:)}^{(3)} \times_1 \dots \times_1 \right. \right. \\ & U_{1(i_{1(c_1-1):})}^{(c_1)} \left. \right) \times_1^2 \left(U_2^{(1)} \times_1 U_{2(i_{21}:)}^{(2)} \times_1 U_{2(i_{22}:)}^{(3)} \times_1 \dots \right. \\ & \times_1 U_{2(i_{2(c_2-1):})}^{(c_2)} \left. \right) \times_1^3 \dots \times_1^N \left(U_N^{(1)} \times_1 U_{N(i_{N1}:)}^{(2)} \times_1 \right. \\ & U_{N(i_{N2}:)}^{(3)} \times_1 \dots \times_1 U_{N(i_{N(c_N-1):})}^{(c_N)} \left. \right) \\ & - t_{i_{11}i_{12}\dots i_{1(c_1-1)}i_{21}i_{22}\dots i_{2(c_2-1)}\dots i_{N1}i_{N2}\dots i_{N(c_N-1)}} \\ & \left(\left(U_1^{(c_1-)} \times_1 U_1^{(c_1)} \right) \circ \left(U_2^{(c_2-)} \times_1 U_2^{(c_2)} \right) \circ \dots \circ \right. \\ & \left. \left(U_N^{(c_N-)} \times_1 U_N^{(c_N)} \right) \right) \end{aligned} \quad (4)$$

$$\begin{aligned} \nabla_{U_k^{(j)}} L = & \left(\underline{C} \times_1^1 \left(U_1^{(1)} \times_1 U_{1(i_{11}:)}^{(2)} \times_1 U_{1(i_{12}:)}^{(3)} \times_1 \dots \times_1 U_{1(i_{1(c_1-1):})}^{(c_1)} \right) \right. \\ & \times_1^2 \left(U_2^{(1)} \times_1 U_{2(i_{21}:)}^{(2)} \times_1 U_{2(i_{22}:)}^{(3)} \times_1 \dots \times_1 U_{2(i_{2(c_2-1):})}^{(c_2)} \right) \\ & \times_1^3 \dots \times_1^N \left(U_N^{(1)} \times_1 U_{N(i_{N1}:)}^{(2)} \times_1 \right. \\ & U_{N(i_{N2}:)}^{(3)} \times_1 \dots \times_1 U_{N(i_{N(c_N-1):})}^{(c_N)} \left. \right) \\ & - t_{i_{11}i_{12}\dots i_{1(c_1-1)}i_{21}i_{22}\dots i_{2(c_2-1)}\dots i_{N1}i_{N2}\dots i_{N(c_N-1)}} \\ & \left(U_k^{(j-)} \circ \left(\underline{C} \times_1^1 \left(U_1^{(1)} \times_1 U_{1(i_{11}:)}^{(2)} \times_1 U_{1(i_{12}:)}^{(3)} \times_1 \dots \times_1 \right. \right. \right. \\ & U_{1(i_{1(c_1-1):})}^{(c_1)} \left. \right) \times_1^2 \left(U_2^{(1)} \times_1 U_{2(i_{21}:)}^{(2)} \times_1 U_{2(i_{22}:)}^{(3)} \times_1 \right. \\ & \dots \times_1 U_{2(i_{2(c_2-1):})}^{(c_2)} \left. \right) \times_1^3 \dots \times_1^{k-1} \\ & \left. \left(U_{(k-1)}^{(1)} \times_1 U_{(k-1)(i_{(k-1)1}:)}^{(2)} \times_1 U_{(k-1)(i_{(k-1)2}:)}^{(3)} \times_1 \dots \times_1 \right. \right. \end{aligned}$$

$$\begin{aligned}
& U_{(k-1)(:i_{(k-1)(c_{k-1}-1):})}^{(c_{k-1})}) \times_1^{k+1} \left(U_{(k+1)}^{(1)} \times_1 U_{(k+1)(:i_{(k+1)1}:)}^{(2)} \right. \\
& \times_1 U_{(k+1)(:i_{(k+1)2}:)}^{(3)} \times_1 \cdots \times_1 U_{(k+1)(:i_{(k+1)(c_{k+1}-1):})}^{(c_{k+1})}) \left. \right) \\
& \times_1^k U_k^{(j+)} \Big), \quad (k = 1, 2, \dots, N, 0 < j \leq c_k) \quad (5)
\end{aligned}$$

where $U_k^{(j-)}$ and $U_k^{(j+)}$ are gotten by the following equations:

$$U_k^{(j-)} = \begin{cases} 1, j = 1 \\ U_k^{(1)} \times_1 U_{k(:i_{k1}:)}^{(2)} \times_1 U_{k(:i_{k2}:)}^{(3)} \times_1 \cdots \\ \times_1 U_{k(:i_{k(j-2):})}^{(j-1)}, 1 < j \leq c_k \end{cases} \quad (6)$$

$$\left[\left[U_k^{(j+)} \right] \right] = \begin{cases} \left[\left[1 \right] \right], j = c_k \\ \left[\left[U_{k(:i_{kj}:)}^{(j+1)} \right] \right] \tilde{\times}_1 \left[\left[U_{k(:i_{k(j+1):})}^{(j+2)} \right] \right] \tilde{\times}_1 \\ \cdots \tilde{\times}_1 \left[\left[U_{k(:i_{k(c_k-1):})}^{(c_k)} \right] \right], 1 \leq j < c_k \end{cases} \quad (7)$$

2) Privacy-Preserving Tucker Decomposition Based on TT:

The goal of privacy-preserving tucker decomposition scheme based on TT is to take advantage of fog-cloud computing to process tucker decomposition based on TT without revealing any information to fog and cloud. The overall steps involved in the proposed scheme are reported in Algorithm 2.

First, the user transforms the tensor $\underline{T}$ to get a higher order tensor T according to the dimensions $I_{11}, I_{12}, \dots, I_{1(c_1-1)}, I_{21}, I_{22}, \dots, I_{2(c_2-1)}, \dots, I_{N1}, I_{N2}, \dots, I_{N(c_N-1)}$ in step 2.

Then, the user generates a random tensor $\underline{D} \in R^{I_{11} \times I_{12} \times \cdots \times I_{1(c_1-1)} \times \cdots \times I_{N1} \times I_{N2} \times \cdots \times I_{N(c_N-1)}}$, and employs perturbation encryption to encrypt the tensor T to get ciphertext $[\underline{T}]$, including $[\underline{T}]_D = \underline{D}$ and $[\underline{T}]_S$ in steps 3 and 4. In step 5, the user sends $\underline{D}$ and $[\underline{T}]_S$ to different clouds and fogs.

After this, fog and cloud transform $[\underline{T}]$ to its encrypted tensor $[\underline{T}]$ in step 7. Fog and cloud initialize the TT cores $U_k^{(1)} \in R^{1 \times R_k \times r_{k1}}; U_k^{(j)} \in R^{r_{k(j-1)} \times I_{k(j-1)} \times r_{kj}}, 1 < j < c_k; U_k^{(c_k)} \in R^{r_{k(c_k-1)} \times I_{k(c_k-1)} \times 1}, \underline{C} \in R^{R_1 \times R_2 \times \cdots \times R_N}$, and encrypt them. Fog and cloud set step size η .

When $t < \text{Iter}_{\max}$ and $L_t > \varepsilon$, fog and cloud perform the following steps. For each $t_{i_{11}i_{12}\cdots i_{1(c_1-1)}i_{21}i_{22}\cdots i_{2(c_2-1)}\cdots i_{N1}i_{N2}\cdots i_{N(c_N-1)}}$, fog and cloud compute $[\nabla_{\underline{C}} L]$, compute $[\nabla_{U_k^{(j)}} L]$

for each core $U_k^{(j)} (k = 1, 2, \dots, N, 0 < j \leq c_k)$, get $[\underline{C}] = [\underline{C}] \tilde{[\eta]} \tilde{[\nabla_{\underline{C}} L]}$, and get $[\nabla_{U_k^{(j)}} L] = [\nabla_{U_k^{(j)}} L] \tilde{[\eta]} \tilde{[\nabla_{U_k^{(j)}} L]}$ for $k = 1, 2, \dots, N, 0 < j \leq c_k$.

After this, fog and cloud transform encrypted tensor $[\underline{C}]$, $[U_k^{(j)}] (k = 1, 2, \dots, N, 0 < j \leq c_k)$ to $[\underline{C}]$, $[U_k^{(j)}] (k = 1, 2, \dots, N, 0 < j \leq c_k)$, and send them to user.

Finally, the user decrypts $[\underline{C}]$, $[U_k^{(j)}] (k = 1, 2, \dots, N, 0 < j \leq c_k)$.

Algorithm 2: Privacy-Preserving Tucker Decomposition Based on Tensor Train.

Input: Tensor $\underline{T} \in R^{I_1 \times I_2 \times \cdots \times I_N}$, maximum number of iterations $\text{Iter}_{\max}$, error threshold ε ;

The dimensions $R_1, R_2, \dots, R_N$ of core tensor $\underline{C}$, number $c_1, c_2, \dots, c_N$ of tensor train cores;

Size $I_{11} \times I_{12} \times \cdots \times I_{1(c_1-1)} \times I_{21} \times I_{22} \times \cdots \times I_{2(c_2-1)} \times \cdots \times I_{N1} \times I_{N2} \times \cdots \times I_{N(c_N-1)}$ transformation for tensor $\underline{T}$, ranks

$r_{11}, r_{12}, \dots, r_{1(c_1-1)}, r_{21}, r_{22}, \dots, r_{2(c_2-1)}, \dots, r_{N1}, r_{N2}, \dots, r_{N(c_N-1)}$ of tensor train cores.

Output: Core tensor $\underline{C} \in R^{R_1 \times R_2 \times \cdots \times R_N}$, N tensor trains $\bar{U}_1, \bar{U}_2, \dots, \bar{U}_N$, where every tensor train $\bar{U}_k = U_k^{(1)} \times_1 U_k^{(2)} \times_k \cdots \times_1 U_k^{(c_k)}$.

- 1: User Side:
- 2: Transform tensor $\underline{T}$ to get tensor T according to dimensions;
- 3: Generate a random tensor $\underline{D} \in R^{I_{11} \times I_{12} \times \cdots \times I_{1(c_1-1)} \times \cdots \times I_{N1} \times I_{N2} \times \cdots \times I_{N(c_N-1)}}$;
- 4: The ciphertext $[\underline{T}]$ of the tensor T is gotten by using perturbation encryption, where $[\underline{T}]$ is composed of $[\underline{T}]_D = \underline{D}$ and $[\underline{T}]_S$;
- 5: Send $\underline{D}$ and $[\underline{T}]_S$ to clouds C_1, C_2 and fogs $F_1, F_2, \dots, F_n$;
- 6: Fog-Cloud Computing:
- 7: Transform $[\underline{T}]$ to its encrypted tensor $[\underline{T}]$;
- 8: Initialize the tensor train cores $U_k^{(1)} \in R^{1 \times R_k \times r_{k1}}; U_k^{(j)} \in R^{r_{k(j-1)} \times I_{k(j-1)} \times r_{kj}}, 1 < j < c_k; U_k^{(c_k)} \in R^{r_{k(c_k-1)} \times I_{k(c_k-1)} \times 1}$ and encrypt them;
- 9: Initialize $\underline{C} \in R^{R_1 \times R_2 \times \cdots \times R_N}$ and encrypt it;
- 10: Set step size η ;
- 11: **while** $t < \text{Iter}_{\max}$ and $L_t > \varepsilon$ **do**
- 12: **for each** $t_{i_{11}i_{12}\cdots i_{1(c_1-1)}i_{21}i_{22}\cdots i_{2(c_2-1)}\cdots i_{N1}i_{N2}\cdots i_{N(c_N-1)}}$ **do**
- 13: Compute $[\nabla_{\underline{C}} L]$;
- 14: Compute $[\nabla_{U_k^{(j)}} L]$ for each core $U_k^{(j)} (k = 1, 2, \dots, N, 0 < j \leq c_k)$;
- 15: $[\underline{C}] = [\underline{C}] \tilde{[\eta]} \tilde{[\nabla_{\underline{C}} L]}$;
- 16: $[\nabla_{U_k^{(j)}} L] = [\nabla_{U_k^{(j)}} L] \tilde{[\eta]} \tilde{[\nabla_{U_k^{(j)}} L]}$ for $k = 1, 2, \dots, N, 0 < j \leq c_k$;
- 17: $t = t + 1$;
- 18: **end for**
- 19: **end while**
- 20: Transform encrypted tensor $[\underline{C}]$, $[U_k^{(j)}] (k = 1, 2, \dots, N, 0 < j \leq c_k)$ to $[\underline{C}]$, $[U_k^{(j)}] (k = 1, 2, \dots, N, 0 < j \leq c_k)$, and send them to user;
- 21: User Side:
- 22: Decrypt $[\underline{C}]$, $[U_k^{(j)}] (k = 1, 2, \dots, N, 0 < j \leq c_k)$.

The encrypted gradients for variables $U_{k(i_{k(j-1):})}^{(j)}$, $\underline{C}$ are derived as follows:

$$\begin{aligned}
& \left[\left[\nabla_{U_{k(i_{k(j-1):})}^{(j)}} L \right] \right] \\
&= \left(\left[\underline{C} \right] \tilde{\times}_1 \left(\left[\left[U_1^{(1)} \right] \right] \tilde{\times}_1 \left[\left[U_{1(i_{11}:})}^{(2)} \right] \right] \tilde{\times}_1 \right. \right. \\
&\quad \left[\left[U_{1(i_{12}:})}^{(3)} \right] \right] \tilde{\times}_1 \cdots \tilde{\times}_1 \left[\left[U_{1(i_{1(c_1-1):})}^{(c_1)} \right] \right] \left. \right) \tilde{\times}_1^2 \left(\left[\left[U_2^{(1)} \right] \right] \right. \\
&\quad \left. \tilde{\times}_1 \left[\left[U_{2(i_{21}:})}^{(2)} \right] \right] \tilde{\times}_1 \left[\left[U_{2(i_{22}:})}^{(3)} \right] \right] \tilde{\times}_1 \cdots \tilde{\times}_1 \left[\left[U_{2(i_{2(c_2-1):})}^{(c_2)} \right] \right] \right) \\
&\quad \tilde{\times}_1^3 \cdots \tilde{\times}_1^N \left(\left[\left[U_N^{(1)} \right] \right] \tilde{\times}_1 \left[\left[U_{N(i_{N1}:})}^{(2)} \right] \right] \tilde{\times}_1 \left[\left[U_{N(i_{N2}:})}^{(3)} \right] \right] \right. \\
&\quad \left. \tilde{\times}_1 \cdots \tilde{\times}_1 \left[\left[U_{N(i_{N(c_N-1):})}^{(c_N)} \right] \right] \right) \\
&\quad \tilde{\times} \left[\left[t_{i_{11}i_{12}\cdots i_{1(c_1-1)}i_{21}i_{22}\cdots i_{2(c_2-1)}\cdots i_{N1}i_{N2}\cdots i_{N(c_N-1)}} \right] \right] \tilde{\times} \\
&\quad \left(\left[\left[U_k^{(j-)} \right] \right] \tilde{\circ} \left(\left[\underline{C} \right] \tilde{\times}_1 \left(\left[\left[U_1^{(1)} \right] \right] \tilde{\times}_1 \left[\left[U_{1(i_{11}:})}^{(2)} \right] \right] \tilde{\times}_1 \right. \right. \\
&\quad \left[\left[U_{1(i_{12}:})}^{(3)} \right] \right] \tilde{\times}_1 \cdots \tilde{\times}_1 \left[\left[U_{1(i_{1(c_1-1):})}^{(c_1)} \right] \right] \left. \right) \tilde{\times}_1^2 \left(\left[\left[U_2^{(1)} \right] \right] \right. \\
&\quad \left. \tilde{\times}_1 \left[\left[U_{2(i_{21}:})}^{(2)} \right] \right] \tilde{\times}_1 \left[\left[U_{2(i_{22}:})}^{(3)} \right] \right] \tilde{\times}_1 \cdots \tilde{\times}_1 \right. \\
&\quad \left. \left[\left[U_{2(i_{2(c_2-1):})}^{(c_2)} \right] \right] \right) \tilde{\times}_1^3 \cdots \tilde{\times}_1^{k-1} \left(\left[\left[U_{(k-1)}^{(1)} \right] \right] \tilde{\times}_1 \right. \\
&\quad \left[\left[U_{(k-1)(i_{(k-1)1}:})}^{(2)} \right] \right] \tilde{\times}_1 \left[\left[U_{(k-1)(i_{(k-1)2}:})}^{(3)} \right] \right] \tilde{\times}_1 \cdots \tilde{\times}_1 \\
&\quad \left. \left[\left[U_{(k-1)(i_{(k-1)(c_{(k-1)-1}):})}^{(c_{(k-1)})} \right] \right] \right) \tilde{\times}_1^{k+1} \left(\left[\left[U_{(k+1)}^{(1)} \right] \right] \tilde{\times}_1 \right. \\
&\quad \left[\left[U_{(k+1)(i_{(k+1)1}:})}^{(2)} \right] \right] \tilde{\times}_1 \left[\left[U_{(k+1)(i_{(k+1)2}:})}^{(3)} \right] \right] \tilde{\times}_1 \cdots \tilde{\times}_1 \\
&\quad \left. \left[\left[U_{(k+1)(i_{(k+1)(c_{(k+1)-1}):})}^{(c_{(k+1)})} \right] \right] \right) \tilde{\times}_1^k \left[\left[U_k^{(j+)} \right] \right] \right) \Big) \\
&\quad (k = 1, 2, \dots, N, 0 < j \leq c_k) \tag{8}
\end{aligned}$$

$$\begin{aligned}
& \left[\left[\nabla_{\underline{C}} L \right] \right] \\
&= \left(\left[\underline{C} \right] \tilde{\times}_1 \left(\left[\left[U_1^{(1)} \right] \right] \tilde{\times}_1 \left[\left[U_{1(i_{11}:})}^{(2)} \right] \right] \tilde{\times}_1 \right. \right. \\
&\quad \left[\left[U_{1(i_{12}:})}^{(3)} \right] \right] \tilde{\times}_1 \cdots \tilde{\times}_1 \left[\left[U_{1(i_{1(c_1-1):})}^{(c_1)} \right] \right] \left. \right) \tilde{\times}_1^2 \left(\left[\left[U_2^{(1)} \right] \right] \right. \\
&\quad \left. \tilde{\times}_1 \left[\left[U_{2(i_{21}:})}^{(2)} \right] \right] \tilde{\times}_1 \left[\left[U_{2(i_{22}:})}^{(3)} \right] \right] \tilde{\times}_1 \cdots \tilde{\times}_1 \right. \\
&\quad \left. \left[\left[U_{2(i_{2(c_2-1):})}^{(c_2)} \right] \right] \right) \tilde{\times}_1^3 \cdots \tilde{\times}_1^N \left(\left[\left[U_N^{(1)} \right] \right] \tilde{\times}_1 \left[\left[U_{N(i_{N1}:})}^{(2)} \right] \right] \right. \\
&\quad \left. \tilde{\times}_1 \left[\left[U_{N(i_{N2}:})}^{(3)} \right] \right] \tilde{\times}_1 \cdots \tilde{\times}_1 \left[\left[U_{N(i_{N(c_N-1):})}^{(c_N)} \right] \right] \right) \\
&\quad \tilde{\times} \left[\left[t_{i_{11}i_{12}\cdots i_{1(c_1-1)}i_{21}i_{22}\cdots i_{2(c_2-1)}\cdots i_{N1}i_{N2}\cdots i_{N(c_N-1)}} \right] \right] \tilde{\times} \\
&\quad \left(\left(\left[\left[U_1^{(c_1-)} \right] \right] \tilde{\times}_1 \left[\left[U_1^{(c_1)} \right] \right] \right) \tilde{\circ} \left(\left[\left[U_2^{(c_2-)} \right] \right] \tilde{\times}_1 \right. \right. \\
&\quad \left. \left[\left[U_2^{(c_2)} \right] \right] \right) \tilde{\circ} \cdots \tilde{\circ} \left(\left[\left[U_N^{(c_N-)} \right] \right] \tilde{\times}_1 \left[\left[U_N^{(c_N)} \right] \right] \right) \Big) \tag{9}
\end{aligned}$$

where $\left[\left[U_k^{(j-)} \right] \right]$ and $\left[\left[U_k^{(j+)} \right] \right]$ are gotten by the following equations:

$$\left[\left[U_k^{(j-)} \right] \right] = \begin{cases} \left[\left[1 \right] \right], j = 1 \\ \left[\left[U_k^{(1)} \right] \right] \tilde{\times}_1 \left[\left[U_{k(i_{k1}:})}^{(2)} \right] \right] \tilde{\times}_1 \left[\left[U_{k(i_{k2}:})}^{(3)} \right] \right] \tilde{\times}_1 \\ \cdots \tilde{\times}_1 \left[\left[U_{k(i_{k(j-2):})}^{(j-1)} \right] \right], 1 < j \leq c_k \end{cases} \tag{10}$$

$$\left[\left[U_k^{(j+)} \right] \right] = \begin{cases} \left[\left[1 \right] \right], j = c_k \\ \left[\left[U_{k(i_{kj}:})}^{(j+1)} \right] \right] \times_1 \left[\left[U_{k(i_{k(j+1):})}^{(j+2)} \right] \right] \times_1 \\ \cdots \times_1 \left[\left[U_{k(i_{k(c_k-1):})}^{(c_k)} \right] \right], 1 \leq j < c_k \end{cases} \tag{11}$$

In our scheme, the user does not participate in the computations of PPTTD after outsourcing the encrypted tensor to fogs $F_1, F_2, \dots, F_n$ and clouds C_1 and C_2 . The fogs $F_1, F_2, \dots, F_n$ and the cloud C_1 only obtain encrypted values, which are gotten by using the homomorphic cryptosystem. Therefore, the PPTTD preserves the data owner's privacy from the fogs $F_1, F_2, \dots, F_n$, and the cloud C_1 . The cloud C_2 only gets blinded values or random values. Therefore, the data owner's privacy is also protected from the cloud C_2 in the PPTTD.

In this article, the following formal definition is employed to describe security against semitrusted adversaries, which has been widely accepted in the existing privacy-preserving schemes, such as literatures [17] and [18].

Definition 1: Suppose that x_k and y_k denote the input and the output of protocol F for participant C_k , respectively. Let $EImage(F, C_k)$ represent the execution image of the protocol F for participant C_k . Let $\equiv$ represent computational indistinguishability. Then, protocol F is secure against semitrusted adversaries if $EImage(F, C_k) \equiv SImage(F, C_k)$, where $SImage(F, C_k)$ is the simulated image of the protocol F for participant C_k and can be simulated from x_k and y_k [17], [18].

Theorem 1: The proposed PPTTD scheme is secure in the semitrusted model.

Proof: Suppose that $EImage(PPTTD, C_1)$ and $SImage(PPTTD, C_1)$ are the execution image and simulated image of the cloud C_1 in the PPTTD scheme, respectively. Then, we can get $EImage(PPTTD, C_1) = \{[T]_S, m_1^k | 1 \leq k \leq K\}$, where $[T]_S = T - [T]_D$, and m_1^k denotes the message that the cloud C_1 receives from the cloud C_2 in the PPTTD scheme. Let $E(X)$ denote any element of tensor X . $SImage(PPTTD, C_1) = \{R_s, r_1^k | 1 \leq k \leq K\}$, where R_s is picked random tensor, and $E(R_s) \in Z_N$. Because $[T]_D$ is one random tensor and $E([T]_D) \in Z_N$, we get $[T]_S \equiv R_s$. Since the basic protocols used in this article are secure, we can call the corresponding simulator to simulate m_1^k for $1 \leq k \leq K$. We can get $m_1^k \equiv r_1^k$ for $1 \leq k \leq K$. As a result $EImage(PPTTD, C_1) \equiv SImage(PPTTD, C_1)$. Suppose that $EImage(PPTTD, C_2)$ and $SImage(PPTTD, C_2)$ are the execution image and simulated image of the cloud C_2 . Similarly, we can get $EImage(PPTTD, C_2) \equiv SImage(PPTTD, C_2)$. Suppose that $EImage(PPTTD, F_j)$ and $SImage(PPTTD, F_j)$ are the

execution image and simulated image of the fog F_j . Similarly, we can get $\text{EImage}(\text{PPTTD}, F_j) \equiv \text{SImage}(\text{PPTTD}, F_j)$ for $1 \leq j \leq n$. ■

Consequently, the proposed PPTTD scheme is secure in the semitrusted model.

IV. PERFORMANCE EVALUATION

We evaluate, in this section, the performance of PPTTD based on gradient descent over blockchain-based encrypted IIoT data. We specifically evaluate our method in terms of efficiency and accuracy using a series of extensive experiments.

Each IIoT data provider collects information from all IIoT devices in its own domain and preprocess the data before sending them to the blockchain for storage. Preprocessing entails aggregation, encryption, and formatting data according to the designed transaction structure. Since IIoT data providers have considerable computing capabilities compared to IIoT devices, we simulate our experiments using Windows 10 PCs with Intel(R) Core(TM) (i5-7500 U) processor and 16 GB RAM for both executing the IIoT data provider functions and running the proposed scheme.

Based on the aforementioned experimental platform, this article implements security operations subprotocols involved in the fog–cloud security calculation model, such as tensor secure single-mode multiplication protocol, TT secure subtraction protocol, and tensor secure outer product protocol. Based on these, PPTTD is implemented. At the same time in reference to this model, we also implement privacy-preserving tucker decomposition based on gradient descent for comparison with our proposed method. Finally, we implement the proposed algorithm into a distributed environment, simulate different numbers of fog end nodes, and test the calculation cost of the “fog–cloud computing layer.” The specific experimental results are discussed in detail.

MNIST dataset consists of 8-b gray-scale images with digits ranging from “0” to “9.” Each class has approximately 6000 examples.

COIL 20 dataset comprises 20 objects with each object having 72 images. Each image has a size of 32×32 pixels, with gray levels of 256 per pixel.

AT&T Faces dataset contains images of 40 people with each person having ten different images. Each image has a size of 92×112 pixels, with gray levels of 256 per pixel.

The synthetic datasets include multiple sets of tensor of size $I_1 \times I_2 \times I_3$, which corresponds to the dimension of the i th order. At the same time, we assume that the size of the core tensor is $R_1 \times R_2 \times R_3$, the number of tensor cores is c_1, c_2 , and c_3 , the size of the original tensor after the ascending order $I_{11} \times I_{12} \times I_{21} \times I_{22} \times I_{31} \times I_{32}$, and the rank between the tensor cores $r_{11}, r_{12}, r_{21}, r_{22}, r_{31}, r_{32}$. The specific synthetic datasets are shown in Table I.

One of the main advantages of the fog–cloud security computing model proposed in this article is the use of fog–cloud computation without revealing the user’s private data. In addition, compared with the traditional homomorphic encryption scheme, the method proposed by the cloud node and the fog node

TABLE I
SYNTHETIC DATASETS

	$I_1 \times I_2 \times I_3$	$I_{11} \times I_{12} \times I_{21} \times I_{22} \times I_{31} \times I_{32}$	$R_1 \times R_2 \times R_3$	$r_{11}, r_{12}, r_{21}, r_{22}, r_{31}, r_{32}$
A	16*100*100	4*4*10*10*10	5*6*7	5,5
B	25*100*100	5*5*10*10*10	5*6*7	5,5
C	36*100*100	6*6*10*10*10	5*6*7	5,5
D	49*100*100	7*7*10*10*10	5*6*7	5,5
E	64*100*100	8*8*10*10*10	5*6*7	5,5
F	81*100*100	9*9*10*10*10	5*6*7	5,5
G	100*100*100	10*10*10*10*10	5*6*7	5,5
H	121*100*100	11*11*10*10*10	5*6*7	5,5

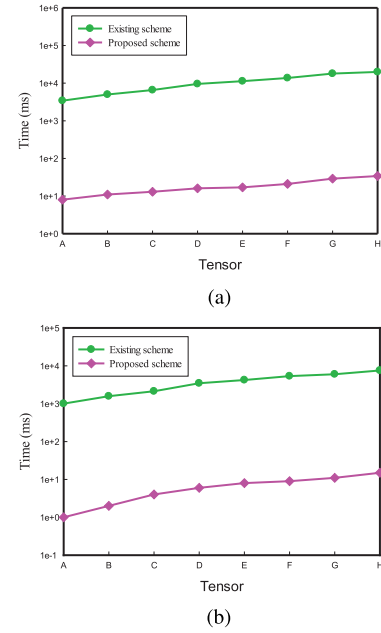


Fig. 1. Time of user sides for the proposed approach and the existing approach. (a) Encryption. (b) Decryption.

to cooperate with each other, this invention greatly reduces the burden on the IIoT devices in theory.

In the PPTTD, the time complexity of the user terminals participating in the calculation is $O(n_e)$ encryption plus $O(n_d)$ decryption. In this article, the computational cost of the scheme and the traditional homomorphic encryption system [5] in the user terminal is compared by the experiments. The experimental results are shown in Fig. 1, where the letter of alphabets in the x -axis represents different tensor sizes and the y -axis represents the user terminal encryption/decryption time. From Fig. 1(a), we can see that the fog–cloud security computing model proposed in this article is much lower in computing cost than the traditional homomorphic encryption system. For example, when the size of the tensor is $121 \times 100 \times 100$ in the fog–cloud security, the time required for the data provider to encrypt the data is 34 ms, whereas the time required for the traditional homomorphic encryption system reaches 19 924 ms. Similarly, in Fig. 1(b), the time required to decrypt the tensor $121 \times 100 \times 100$ by the two schemes is 15 and 7558 ms, respectively. Therefore, the experimental results show that the proposed scheme has

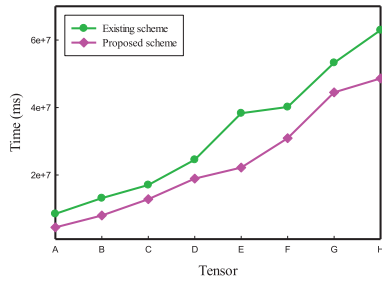


Fig. 2. Comparison of PPTTD and privacy-preserving tucker decomposition based on gradient descent over execution time.

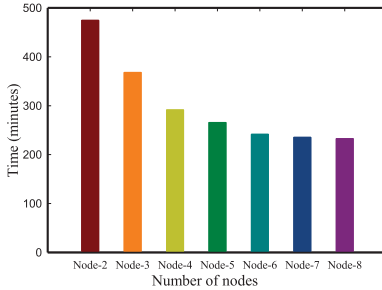


Fig. 3. Time cost of the proposed tucker decomposition based on TT approach for varying number of fog nodes.

less computational cost in the user terminal, so it can better adapt to the user terminal equipment with different processing capabilities.

In order to verify the computational efficiency of the PPTTD proposed in this article, we implement the privacy-preserving tucker decomposition based on gradient descent on the fog–cloud security computing model and its subprotocols. Through the implementation of these two algorithms, we have the following experimental results.

Fig. 2 shows the execution time of the two algorithms in the cloud and fog. As can be seen from the figure, the PPTTD proposed in this article has obvious advantages in the execution time. For example, when the amount of data is $64 \times 100 \times 100$, the average time required for the traditional tucker decomposition based on gradient descent to execute in the cloud and fog is 3.83×10^7 ms, and the average time required for the tucker decomposition based on TT proposed in this article is 2.22×10^7 ms, which is about 42% faster. This shows that the algorithm has a significant improvement in time efficiency.

As shown in Fig. 3, when we gradually increase the number of fog end nodes from 1 to 8, the distributed PPTTD has a significant reduction in the calculation time of the fog–cloud computing layer. For example, when the fog nodes are increased to four, the execution time of the privacy-preserving algorithm is 291 min, which is 38.6% less than the time when the number of fog nodes is 2. From the experimental results, the total execution time can further be reduced by increasing the number of fog nodes.

We perform the classification using the results of our method. We evaluate the accuracy of our PPTTD on real datasets. Fig. 4 shows the classification accuracies of the method using our privacy-preserving tucker decomposition and the classification

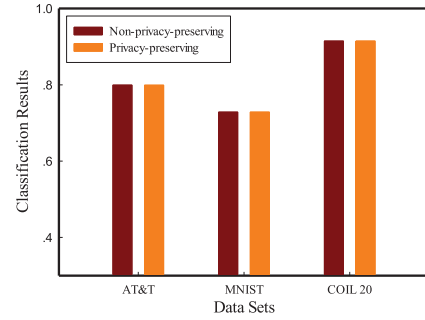


Fig. 4. Classification accuracies on real datasets.

accuracies of the traditional method. In the experiments, the classification accuracies of our method are 79.92%, 72.87%, and 91.47% on AT&T Faces dataset, MNIST dataset, and COIL 20 dataset, respectively. We observe that the classification accuracies of our privacy-preserving method are nearly equal to those of the traditional method, which demonstrates the effectiveness of the proposed method.

V. RELATED WORK

A. Privacy-Preserving Methods

There are quite a number of techniques that have been used across many fields to protect the privacy of data [1]. One such technique is data anonymization [19], a method that attempts to protect personal and sensitive information from being easily identified. This method, however, is prone to numerous types of attacks especially if the attacker has background knowledge.

Another method for protecting data and user privacy is DP [20], a technique that ensures data privacy by perturbing the data or adding noise to the computational process before the data are shared. The technique allows multiple parties to compute a joint function without revealing their sensitive input information. DP-based techniques usually achieves higher accuracy in protecting data privacy at the expense of reducing the quality of data, thus analysis based on the perturbed data can lead to inaccurate results.

To achieve better guarantee in data privacy protection, homomorphic cryptosystem is used to compute on encrypted data. Homomorphic encryption allows direct computation on encrypted data while making sure that the computations are correct. Several homomorphic encryption based algorithms have been proposed in literatures, i.e., [5].

B. Blockchain for IoT

There has been ongoing research on how to make IoT blockchain-enabled to improve IoT systems' privacy [21], [22]. Many of the previous researches have focused on access control, secure applications, cloud data storage, and cloud computations. The introduction of Bitcoin brought about numerous possibilities on how to use blockchain technology in both financial and nonfinancial applications. In the recent past, there has been several attempts to use the blockchain technology in nonfinancial

applications [10], [23]. It is thus not difficult to see how this technology can easily be applicable in many big data applications, including but not limited to cyber-physical-social systems (CPSS), IIoT, and smart cities.

Ouaddah *et al.* [24] and Rahulamathavan *et al.* [25] focused on authentication as a way of securing IoT systems. Ouaddah *et al.* [24], in particular, proposed a framework for access control based on blockchain technology. By using this decentralized framework, users have control of their data. Rahulamathavan *et al.* [25] proposed a privacy-preserving private transaction sharing platform for IoT applications using blockchain and attribute based encryption techniques. In their framework, only users with certain allowed attributes are allowed to use the stored data. Zyskind *et al.* [26] implemented a user data management scheme built on the blockchain utilizing off-chain storage for privacy. He *et al.* [27] proposed a privacy-preserving IoT device management framework using blockchain that provide efficient attribute-based and time-bound access supporting automatic key revocation.

VI. CONCLUSION

In this article, we presented a novel PPTTD-based on gradient descent over blockchain-based encrypted IIoT data, which handles data privacy challenges by using blockchain techniques. We utilized secure TT-based tucker decomposition to process blockchain-based encrypted data collected from various IIoT environments. In our approach, homomorphic encryption was used to construct the secure TT-based tucker decomposition. We finally demonstrated from experiments the security and efficiency of our proposed approach.

Part of our future work is to develop a general framework for implementing different types of privacy-preserving tensor decomposition using data from multiple IIoT.

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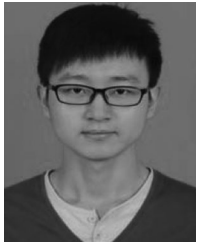
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